

Artificial intelligence in dizziness and vertigo: A review of machine learning and deep learning applications for multimodal diagnosis and management

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ABSTRACT

Dizziness/vertigo is among the most common symptoms in clinical practice, affecting approximately 15%–20% of adults each year. Its etiologies are complex and involve multiple systems, including the central nervous system, peripheral vestibular system, and psychological factors. Because symptoms are highly subjective and key physical signs are difficult to capture, diagnosis remains heavily dependent on clinicians' experience, particularly in emergency settings, where central causes such as posterior circulation stroke are at risk of being missed. Artificial intelligence (AI), especially machine learning (ML) and deep learning (DL), can automatically extract latent features from high-dimensional and multimodal clinical data, providing a new approach for the diagnosis and management of dizziness/vertigo disorders. This review summarizes recent advances in AI applications for dizziness/vertigo disorders, covering multimodal data sources such as medical imaging, videonystagmography (VNG), video head impulse test (vHIT), postural and gait assessment, and structured clinical information. It also summarizes AI applications in benign paroxysmal positional vertigo, Ménière's disease, vestibular migraine, posterior circulation stroke, orthostatic intolerance, and related disorders. Current studies have shown promising performance of AI in multiple scenarios. For example, a DL-based nystagmus pattern classification model achieved an accuracy of 94.91% using VNG videos; an LSTM model based on raw vHIT time-series data achieved an accuracy of 87.9% for stroke detection in acute vestibular syndrome; and a CNN-based automated inner ear MRI segmentation model enabled objective quantification of endolymphatic hydrops. However, current studies are still limited by small sample sizes, a high proportion of single-center datasets, inconsistent annotation, insufficient model interpretability, and a lack of prospective validation, which restrict their clinical translation. Future research should strengthen the development of standardized multicenter datasets, advance explainable AI, federated learning, and multimodal large models, and promote the expansion of AI from diagnostic assistance to treatment decision support and long-term management.

1. Introduction

Dizziness/vertigo are extremely common symptoms in clinical practice. Epidemiological data indicate that approximately 15%–20% of adults experience dizziness/vertigo each year, and the incidence increases with age, reaching up to 30% among individuals older than 60 years.^{1,2} The etiologies are heterogeneous, ranging from peripheral vestibular disorders (e.g., benign paroxysmal positional vertigo and Ménière's disease) to central nervous system conditions (e.g., vestibular migraine and posterior circulation stroke). Early and accurate differentiation between peripheral and central causes is crucial for reducing the risk of disability and mortality.

However, the diagnosis of dizziness/vertigo remains challenging due to the wide spectrum of clinical manifestations, vague and variable symptom descriptions, and the difficulty of capturing key physical signs. Moreover, diagnostic decision-making largely depends on clinicians' expertise and accumulated experience, particularly in interpreting complex oculomotor examinations and specific provocation tests. Inexperienced physicians may miss critical findings or make incorrect judgments. Although vestibular function tests can provide objective evidence, their widespread implementation is limited by high cost, operational complexity, time consumption, and, in some cases, ongoing controversies regarding interpretation criteria. Collectively, these limitations highlight the urgent need to improve both the

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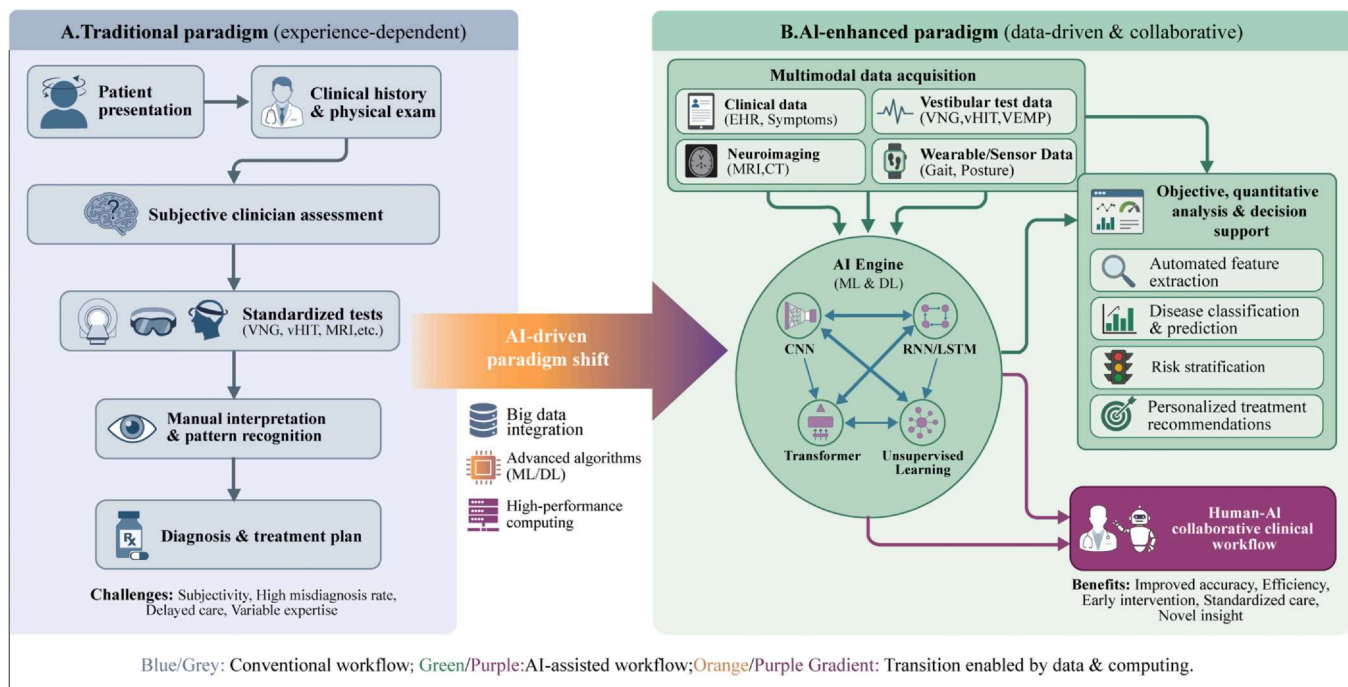


Fig. 1. AI-enabled paradigm shift in dizziness/vertigo diagnosis and management. VNG, videonystagmography; vHIT, video head impulse test; VEMP, vestibular evoked myogenic potential; MRI, magnetic resonance imaging; CT, computed tomography; AI, artificial intelligence; EHR, electronic health record; ML, machine learning; DL, deep learning; CNN, convolutional neural network; RNN, recurrent neural network; LSTM, long short-term memory.

accuracy and efficiency of traditional diagnostic approaches for dizziness/vertigo.

In recent years, the rapid advancement of artificial intelligence (AI) in medicine has created new opportunities for improving the diagnosis of dizziness/vertigo. A key strength of AI lies in its ability to automatically learn and identify clinically meaningful patterns from large-scale, high-dimensional, and complex datasets. By leveraging AI algorithms, multimodal patient information—including symptoms, physical signs, and examination results—can be rapidly analyzed and classified, thereby assisting clinicians in achieving faster and more accurate diagnoses. Against this background, this review summarizes the latest research progress on the application of AI in the diagnosis and management of dizziness/vertigo disorders, with a focus on AI-based multimodal assessments, disease classification, and differential diagnosis, applications in specific disorders, as well as current challenges and future perspectives.

2. Overview of AI

AI generally encompasses two major branches: machine learning (ML) and deep learning (DL). Among ML paradigms, supervised learning is the most commonly used approach, in which model parameters are optimized using labeled training data to enable classification or prediction on unseen data.³ For example, by inputting a patient's clinical symptoms, physical signs, demographic characteristics, and standardized test parameters, supervised models can output the predicted probability of specific disease categories. Common supervised learning algorithms include logistic regression, support vector machines (SVM), decision trees, random forest (RF), and gradient boosting methods (e.g., XGBoost and LightGBM). These algorithms have distinct strengths and limitations and are suitable for different types and scales of datasets. In addition, unsupervised learning techniques, such as clustering and dimensionality reduction, can be applied to identify dominant patterns within data and to uncover potential subtypes among patients with dizziness/vertigo.

DL is a subfield of ML that primarily relies on multilayer neural networks to automatically learn hierarchical feature representations

from raw data, thereby enabling pattern recognition and prediction in high-dimensional and nonlinear datasets. In the diagnosis of dizziness/vertigo, DL is particularly well-suited for processing high-dimensional raw signals and imaging data.⁴ Convolutional neural networks (CNN) can efficiently extract spatial structural features and have been widely used for grid-like data, such as images and video frames. Recurrent neural networks (RNN) capture temporal dependencies in sequential data through internal memory units and are commonly applied to time-series signals or natural language text. More recently, Transformer architectures, built upon self-attention mechanisms, can model global dependencies between any positions within a sequence and have demonstrated strong performance in handling long sequences and multimodal data.

Overall, AI is capable of simulating key human cognitive functions, including perception, learning, reasoning, and decision-making. As early as the 20th century, the concept of computer-aided diagnosis was proposed, and early expert systems were developed to support clinical decision-making.⁵ With advances in algorithms and computational power, modern AI has been widely applied in medical image analysis, disease risk prediction, and related fields. Importantly, AI is not intended to replace physicians; rather, it serves as an intelligent assistant to help primary care providers or non-specialists identify warning signs, optimize the selection of diagnostic tests, and ultimately shorten the diagnostic pathway. In summary, integrating AI into the diagnostic workflow for dizziness/vertigo holds promise for improving both diagnostic efficiency and accuracy, thereby facilitating the development of individualized treatment strategies (Fig. 1).

3. Applications of AI in multimodal assessments

3.1. AI-assisted interpretation of imaging findings

Imaging examinations in patients with dizziness/vertigo are primarily used to identify structural abnormalities, such as inner ear lesions or central nervous system pathology. Conventional Magnetic resonance imaging (MRI) and computed tomography (CT) interpretation

largely depends on manual reading, which is subject to interobserver variability and may lead to subtle findings being overlooked. In recent years, radiomics and AI have provided novel approaches for mining high-dimensional information from medical images. In the diagnosis of Ménière's disease (MD), conventional MRI often fails to directly visualize the characteristic endolymphatic hydrops (EH). However, radiomics-based studies have offered new insights. A multicenter retrospective study extracted multiple texture features from routine T2-weighted inner ear MRI and trained a machine learning–based multi-layer perceptron model to differentiate patients with MD from healthy controls. The model achieved an accuracy of 82%, a sensitivity of 83%, and a specificity of 82% in an independent test set.⁶ These findings suggest that radiomics combined with ML may serve as a noninvasive adjunctive diagnostic tool for MD. Cho et al.⁷ developed a CNN-based model, termed *INHEARIT*, to automatically segment the cochlea and vestibule on contrast-enhanced inner ear MRI and to quantify the EH ratio. The EH ratios measured by the AI system showed excellent agreement with manual measurements performed by neurotologists (intraclass correlation coefficient = 0.971), providing the first evidence that quantitative assessment of EH can be fully automated using AI. This automated workflow substantially reduces the time and complexity required for manual MRI interpretation, offering a rapid and accurate tool for objective diagnosis and treatment monitoring in MD. Beyond inner ear imaging, AI applications in brain imaging also deserve attention. For central causes of vertigo (e.g., brainstem or cerebellar infarction), AI can assist in identifying subtle ischemic lesions on neuroimaging. DL has been widely applied to MRI and has demonstrated strong performance in lesion detection and segmentation. Approaches such as 3D U-Net architectures and transfer learning may further improve sensitivity for small-volume and low-contrast lesions. Nevertheless, additional validation and optimization are required to facilitate clinical translation, particularly with respect to multicenter large-scale datasets, model interpretability, and inference efficiency.⁸

3.2. AI-assisted vestibular function testing and oculomotor analysis

Videonystagmography (VNG), the video head impulse test (vHIT), and vestibular evoked myogenic potentials (VEMPs) are essential examinations for evaluating vestibular function. AI can leverage the large amount of physiological signal information embedded in these tests for pattern recognition and classification. Slama et al.⁹ analyzed VNG nystagmus videos by performing automatic pupil segmentation and tracking to extract features such as nystagmus frequency, fast and slow phase velocities, and periodic fluctuations. Feature selection was then performed using Fisher's discriminant criterion, and an SVM model was constructed to distinguish patients with vestibular neuritis from healthy controls, achieving a classification accuracy of approximately 94%. Kong et al.¹⁰ proposed a nystagmus pattern classification framework that combines DL with optical-flow analysis to automatically identify different types of nystagmus from VNG videos. In this framework, LiteFlowNet is first used to extract optical-flow information from consecutive video frames, thereby quantifying changes in eye movement direction, velocity, and amplitude. A ConvNeXt model, which is based on a CNN architecture, is then applied to extract spatial features, while a long short-term memory (LSTM) network is used to integrate the temporal dynamics of nystagmus over time. By jointly analyzing the spatial morphology and temporal evolution of nystagmus, this framework achieved a classification accuracy of 94.91% and substantially improved the detection of torsional nystagmus. However, further validation in larger samples and multicenter datasets is still required. The favorable performance of CNN- and LSTM-based models in VNG video analysis can be attributed mainly to their ability to jointly model spatial features and temporal dynamics. Conventional nystagmus analysis usually relies on manually or semi-automatically extracted predefined parameters, such as slow-phase velocity, fast-phase direction, and duration. Although clinically interpretable, these parameters capture

only selected aspects of nystagmus signals. In contrast, CNN can automatically learn spatial patterns from consecutive video frames or optical-flow maps, including eye movement trajectories, pupil-boundary changes, nystagmus direction, amplitude, and local morphology, making them particularly suitable for detecting subtle differences among horizontal, vertical, and torsional nystagmus.¹¹ LSTM can further model the dynamic evolution of nystagmus over time, including the alternation between fast and slow phases, fluctuations in nystagmus frequency, velocity trends, and temporal dependencies across different phases.¹² This ability to capture both spatial and temporal information represents an important advantage over conventional signal analysis methods based on handcrafted features.

Static ocular torsion is an important objective indicator of vestibular system function, particularly reflecting the functional status of the otolith organs. Quantitative analysis of static ocular torsion is increasingly becoming a key reference for evaluating vestibular function and the vestibular compensation process. One study addressed the challenge of automated detection of static ocular torsion and proposed a CNN-based intelligent analysis approach to quantitatively estimate ocular torsion angles from fundus images. By applying image rotation, nearly 13,000 synthetic images were generated, and transfer learning was further employed to develop ResNet-based deep models for binary and multiclass classification. The model demonstrated good performance in the binary classification task, achieving an accuracy of 92%, while the accuracy in the four-class classification task was 77%. Grad-CAM visualization indicated that the network focused on key anatomical structures consistent with clinical landmarks used for ocular torsion assessment, supporting the interpretability of the model's predictions.¹³ These findings suggest that a synthetic data–driven DL approach can achieve high-accuracy classification of static ocular torsion even in the absence of large-scale annotated datasets, and may have potential clinical value in scenarios such as acute dizziness/vertigo, vestibular disorders, and ocular motor palsy.

Similarly, for vHIT examinations, conventional analysis often reduces complex eye- and head-velocity traces to a single metric, namely the vestibulo-ocular reflex (VOR) gain. Although intuitive, this simplification inevitably discards rich dynamic information embedded in the raw signals. A recent study explored the feasibility of using unprocessed vHIT raw data directly for stroke diagnosis.¹⁴ By applying an LSTM model to the time-series recordings of eye and head velocities during vHIT, the model achieved a stroke diagnostic accuracy of 87.9% (sensitivity 87.7%, specificity 88.4%) among 57 patients with acute vestibular syndrome (AVS). This performance was comparable to that of the traditional VOR gain threshold-based method, demonstrating the ability of DL to extract informative patterns from dynamic VOR data. Mechanistically, conventional VOR gain calculation typically treats each time point, or short time window, independently and then averages the results, thereby ignoring the temporal structure of the VOR response. Central lesions, particularly brainstem or cerebellar strokes, do not always manifest as simple abnormalities in VOR gain. Time-series models such as LSTM can integrate continuous eye and head velocity data, learn dependencies across different time points, and capture nonlinear and high-dimensional dynamic features. These features may include the complete slow-phase VOR profile within a time window, such as the rising phase, peak phase, and falling phase of the velocity curve, as well as fast-phase responses, including compensatory saccades. By learning these temporal patterns, the model may distinguish different pathological states. Therefore, AI-based analysis of raw vHIT signals provides incremental information beyond a single VOR gain metric, enabling the model to identify latent signal patterns associated with central lesions even in patients with apparently normal or borderline VOR gain, thereby improving the sensitivity and objectivity of stroke screening. In addition, AI can integrate multiple vestibular function tests. Previous studies have shown that ML models combining multimodal vestibular data, including VNG, vHIT, VEMPs, and caloric testing, can significantly improve the accuracy of differential diagnosis.¹⁵

3.3. AI-assisted postural and gait assessment

Postural balance assessment typically involves three major sensory systems: vestibular, visual, and proprioceptive inputs. Conventional evaluation methods mainly rely on clinical observation, standardized scales, and static or dynamic balance tests. However, these approaches are, to some extent, influenced by subjective factors and may not fully capture an individual's balance status. AI is increasingly optimizing postural assessment pathways by enabling automated processing and fusion of multisource signals, thereby improving objectivity. Kamogashira et al.¹⁶ applied multiple ML algorithms to center-of-pressure (COP) data obtained from posturography in patients with vertigo to predict vestibular dysfunction, using caloric testing as the reference standard. Among the evaluated models, the gradient-boosted decision tree achieved the best predictive performance, with an AUC of 0.90. Choi et al.¹⁷ proposed a DL-based approach to evaluate postural control in patients with vertigo by using CNNs to analyze spectrograms derived from COP signals during quiet standing to estimate the equilibrium score (ES). Among the four tested CNN architectures (GoogleNet, ResNet–18, SqueezeNet, and VGG16), ResNet–18 performed best. Notably, this method enabled accurate estimation of ES from a single quiet-standing COP recording and quantified the relative contributions of the visual, vestibular, and somatosensory systems, highlighting its potential to simplify balance assessment.

In gait analysis, AI can also predict fall risk from gait parameters collected by wearable sensors. Using the XGBoost algorithm, researchers found that step length and gait speed were important variables for predicting future falls.¹⁸ Zhang et al.¹⁹ assessed gait stability in patients with BPPV using wearable accelerometers and found that BPPV patients adopted a more conservative gait pattern with significantly reduced walking stability. An SVM model based on gait parameters achieved an AUC of 0.78 for distinguishing BPPV patients from healthy controls. Furthermore, the model yielded AUCs of 0.83, 0.85, and 0.96 for classifying mild, moderate, and severe subgroups according to the Dizziness Handicap Inventory (DHI), respectively, demonstrating favorable predictive capability. Collectively, these findings suggest that AI has the potential to objectively quantify balance function in patients with vertigo, enabling vertigo monitoring and early warning for high-risk populations.

4. Applications of AI in the classification of dizziness/vertigo

AI is not limited to analyzing individual diagnostic tests; it can also integrate multisource clinical data to support disease classification and prediction. Moreover, AI-driven approaches may facilitate the discovery of underlying disease mechanisms and previously unrecognized subtypes.

4.1. AI-Assisted differentiation between peripheral and central causes of dizziness/vertigo

Rapid differentiation between central and peripheral causes of dizziness/vertigo is critical for clinical decision-making. Conventional clinical tools, such as the HINTS examination and the ABCD2 score, are widely used to rapidly assess stroke risk. Although HINTS demonstrates high sensitivity and specificity for stroke diagnosis in patients with typical AVS, its accuracy is highly dependent on the examiner's expertise and experience. Moreover, its diagnostic performance decreases substantially in stroke patients who do not present with typical AVS features. AI has emerged as a promising strategy to address these limitations. By integrating richer clinical information, AI can construct more robust and comprehensive diagnostic models than traditional scoring systems.

A prospective study enrolled 108 patients with acute vertigo, including 40 with posterior circulation stroke and 68 with vestibular neuritis, and systematically compared the diagnostic performance of

multiple ML algorithms with that of HINTS and the ABCD2 score. The geometric deep learning model MultiGMC achieved the best performance in distinguishing posterior circulation stroke from vestibular neuritis, with an AUC of 0.96, which was substantially higher than that of HINTS (AUC = 0.71) and ABCD2 (AUC = 0.58). Importantly, when stroke patients did not present with typical AVS features (i.e., without spontaneous nystagmus), the diagnostic performance of HINTS declined sharply (AUC = 0.54), whereas the ML models maintained relatively high diagnostic accuracy.²⁰ Wang et al.²¹ developed ML models based on raw vHIT head and eye velocity traces from patients with acute vestibular syndrome to differentiate posterior circulation stroke from vestibular neuritis. The study compared several time-series algorithms, including Rocket, Arsenal, Catch22, and random interval classifiers. Among these models, Rocket showed the best performance, achieving an accuracy of 87.8% in the external validation cohort. Its diagnostic performance was comparable to that of neurotology specialists and superior to the traditional approach based solely on VOR gain cut-off values. Notably, the test can be completed within minutes, providing emergency physicians with a rapid and objective decision-support tool and offering a potential automated alternative to the HINTS examination. With further validation in multicenter cohorts, such AI-assisted models may be incorporated into emergency triage systems, improving stroke recognition in resource-limited settings and enabling timely referral and treatment for high-risk patients with central dizziness/vertigo.

4.2. AI-assisted multiclass classification of dizziness/vertigo

Beyond the broad task of central–peripheral differentiation, a more common clinical challenge involves complex differential diagnosis caused by coexisting etiologies and overlapping symptoms. In this context, AI has also made substantial progress in multiclass classification of dizziness/vertigo disorders. One study developed an AI-assisted decision-support system that integrates ML with clinical expertise to address the diagnostic complexity of vestibular disorders. The system maintained high sensitivity for common conditions such as BPPV and vestibular migraine (VM), facilitating rapid clinical screening. Meanwhile, it achieved high specificity for disorders requiring more nuanced differentiation, including MD, hemodynamic orthostatic dizziness (HOD), and persistent postural-perceptual dizziness (PPPD), thereby reducing the risk of misdiagnosis. The model incorporated 50 clinical features selected through a combination of algorithmic approaches and expert knowledge, and was trained using the CatBoost gradient-boosted decision tree algorithm. In an independent test set, the overall accuracy reached 88.4%, with 60.9% of cases fully matching the correct diagnosis, 27.5% partially matching, and only 11.6% misclassified. These findings suggest that ML models combined with clinically informed rules can, to some extent, emulate the diagnostic reasoning of specialists.²² Similar multiclass models have also been reported. For example, Vivar et al.²³ developed a tool based on a large vertigo patient database, performing multivariable analyses and training multiple algorithms to classify conditions such as functional dizziness, MD, and VM, achieving favorable classification performance. Collectively, these studies indicate that AI models trained on large-scale datasets may support outpatient diagnostic decision-making for vertigo, particularly for non-specialists, by providing differential diagnostic suggestions and reducing missed diagnoses or misclassification of complex cases.

4.3. AI-assisted disease subtyping and mechanistic exploration

AI is not limited to supervised classification tasks; unsupervised learning has also demonstrated substantial value in exploring disease mechanisms and identifying clinically meaningful subtypes. Visscher et al.²⁴ performed unsupervised clustering analysis of balance and vestibular diagnostic data from 96 patients with concussion. Using a

multilayer self-organizing map algorithm, they identified two patient clusters characterized by distinct vestibular profiles. The maximum slow-phase velocity in caloric testing and dynamic visual acuity were key parameters differentiating the two groups. Notably, one cluster showed significantly higher proportions of headache, blurred vision, balance impairment, and vestibular dysfunction, suggesting that ML can reveal latent concussion subtypes without relying on predefined diagnostic labels and may facilitate a deeper understanding of neuro-functional heterogeneity following concussion. In another study, 244 patients with VM diagnosed according to Bárány Society criteria were included, and clustering analysis was conducted to characterize phenotypic heterogeneity. The results indicated substantial variability in clinical features, particularly in the duration of vertigo attacks, the presence of migraine-associated symptoms, and cochlear manifestations. Five distinct clinical subgroups were identified, highlighting the heterogeneity of VM and suggesting potential biological stratification within this disorder.²⁵

AI can also help identify risk factors and uncover potential mechanisms underlying dizziness/vertigo disorders. Recent studies have attempted to incorporate laboratory tests and biomarkers into predictive models to explore their associations with disease occurrence. For example, in developing an intelligent diagnostic model for BPPV, researchers extended beyond clinical symptoms and included multiple inflammatory and metabolic indicators as candidate features. After feature selection using LASSO regression and RF, neutrophil count, lymphocyte count, C-reactive protein, ferritin level, vitamin D deficiency, and related parameters were identified as independent risk factors for BPPV, whereas monocyte count appeared to be a protective factor. The final multivariable model achieved an AUC of 0.927, demonstrating strong predictive performance. These findings suggest that inflammatory immune responses and metabolic dysregulation may contribute to the pathogenesis of BPPV.²⁶ Overall, AI not only provides diagnostic outputs but can also serve as a powerful research tool to discover meaningful patterns from complex clinical and biological data, thereby improving our understanding of the underlying nature of dizziness/vertigo disorders.

5. AI applications in common dizziness/vertigo

5.1. AI applications in BPPV

BPPV is one of the most common causes of vertigo, accounting for more than one-third of all vertigo cases.²⁶ It is caused by dislodged otoconia entering the semicircular canals and is clinically characterized by brief episodes of position-triggered vertigo. Although BPPV is a benign disorder, accurate diagnosis and effective canalith repositioning maneuvers require substantial clinical expertise. Current applications of AI in BPPV mainly focus on two aspects: assisting diagnosis and optimizing repositioning treatment strategies.

Traditional diagnosis of BPPV relies on observing positional nystagmus during provocation maneuvers such as the Dix–Hallpike test; however, identifying specific subtypes and differentiating BPPV from other positional vertigo disorders often requires specialist expertise. ML models can leverage patients' symptoms, medical history, and examination findings to establish discriminative algorithms. Mun et al.²⁷ developed and evaluated nystagmus detection models based on VNG videos. In their approach, the U-Net was first used for pupil segmentation, and horizontal eye-movement time-series data were then extracted. CNN- and LSTM-based models were subsequently developed for nystagmus detection. Among these models, the 1D-CNN model showed the best performance (accuracy 91.02%, F1-score 92.68%). By integrating pupil-tracking analysis, this approach enabled high-precision eye movement recognition and outperformed previous ML methods. Lu et al.²⁸ developed a multimodal DL model for BPPV by integrating nystagmus video features with head-position information. By incorporating an autoencoder and a cross-attention mechanism, the

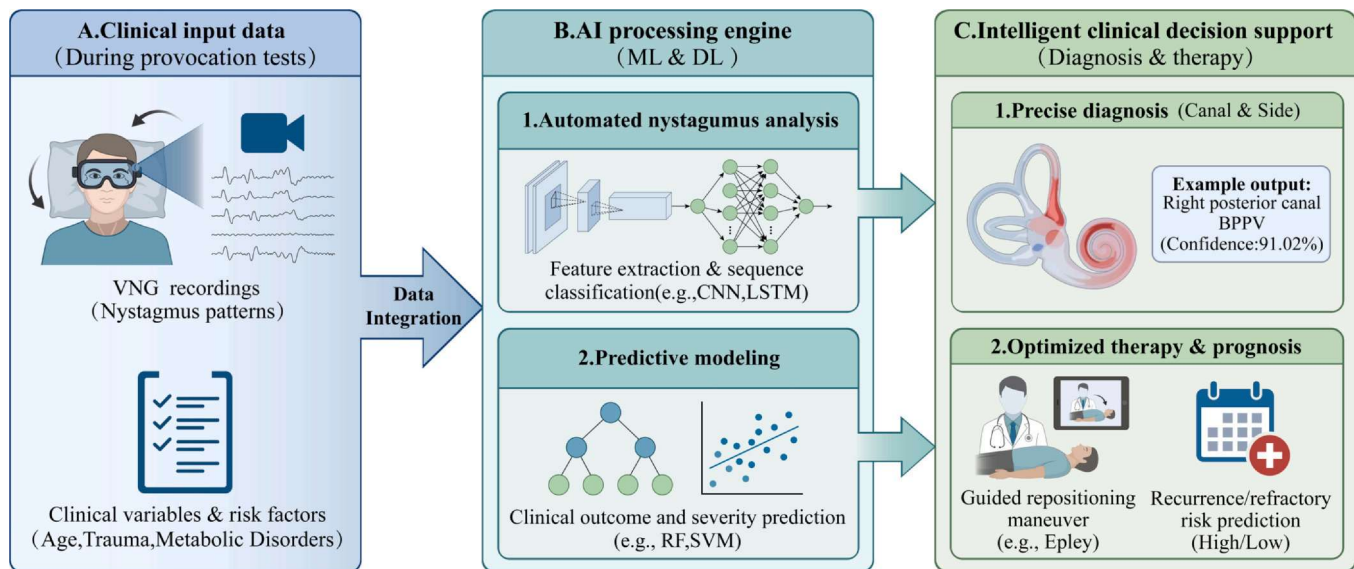
model enabled assisted diagnosis of different BPPV subtypes and achieved a mean accuracy of 81.7%, significantly outperforming conventional time-series classification approaches. This finding highlights the importance of fusing head posture information with nystagmus data for AI-assisted BPPV diagnosis. Collectively, these studies suggest that AI can objectively and rapidly identify characteristic nystagmus patterns in typical BPPV, potentially reducing reliance on subjective clinical experience—particularly benefiting primary care providers and non-specialists. Notably, AI models may also predict the affected semicircular canal subtype of BPPV to some extent based on the Dizziness Handicap Inventory (DHI).²⁹ Overall, the integration of AI is shifting BPPV diagnosis from experience-based qualitative assessment toward quantitatively supported decision-making.

Canalith repositioning maneuvers remain the mainstay treatment for BPPV, and most patients recover after one or two sessions. However, a subset of patients with complicated or refractory BPPV (e.g., multi-canal involvement or frequent recurrence) respond poorly to standard treatment. AI has therefore been explored as a means to optimize repositioning strategies. One study developed an intelligent repositioning system (the SRM-IV diagnostic and treatment system) equipped with AI algorithms. While the patient is seated, the system automatically rotates the chair to predefined positions to provoke nystagmus, and a camera continuously monitors nystagmus parameters in real time. Based on these data, the AI algorithm determines the affected ear and involved semicircular canal and then selects the optimal repositioning maneuver. Compared with conventional manual repositioning, SRM-IV significantly improved diagnostic efficiency in refractory BPPV, achieving a sensitivity of 92.3% and a specificity of 89.7%.³⁰ In addition, ML has been used to predict the number of repositioning sessions required for BPPV patients. Baydan-Aran et al. applied a random forest model to clinical variables to identify patients who might require repeated repositioning, and the predictive accuracy for those needing ≥ 3 maneuvers was markedly improved.³¹ This may help clinicians recognize difficult cases early and adjust follow-up and rehabilitation strategies accordingly. Overall, AI not only supports the diagnosis of BPPV but is also gradually being integrated into therapeutic workflows, enabling human-machine collaboration to improve both efficacy and efficiency (Fig. 2).

5.2. AI applications in MD

MD is an idiopathic inner ear disorder characterized by recurrent spontaneous vertigo, fluctuating hearing loss, tinnitus, and aural fullness. Its pathological hallmark is EH. Current AI applications in MD mainly focus on assisting differential diagnosis and enabling objective quantification of disease-related biomarkers. Wang et al.¹⁵ integrated medical history, nystagmus findings, audiological data, and vestibular function test results using ML models to differentiate MD from VM with high accuracy. In a neurotology clinic scenario where comprehensive examination data were available, both AdaBoost and random forest achieved an accuracy of up to 97.8%. Even in a general outpatient setting with only history-based information, the model maintained an accuracy exceeding 92%. Importantly, feature contribution analysis indicated that several history-related elements (e.g., auditory symptoms) were among the most influential factors for distinguishing MD from VM, suggesting that AI can capture subtle yet informative feature combinations that may be underutilized in routine clinical reasoning and thereby address a long-standing challenge in differential diagnosis. Another major difficulty lies in obtaining objective evidence to confirm MD. Although the clinical diagnosis is primarily based on Bárány Society criteria, AI may provide indirect supportive evidence. For instance, as described above, radiomics-based approaches have demonstrated the ability to distinguish MD from healthy controls using MRI-derived texture features.⁶

Given that EH is the distinctive pathophysiological hallmark of MD, objective quantification of hydrops severity has become a major



AI integrates complex VNG and clinical data to automate BPPV diagnosis and provide personalized therapeutic guidance, improving efficiency and outcomes.

Fig. 2. AI-driven workflow for precise diagnosis and management of BPPV. VNG, videonystagmography; BPPV, benign paroxysmal positional vertigo; AI, artificial intelligence; ML, machine learning; DL, deep learning; CNN, convolutional neural network; LSTM, long short-term memory; RF, random forest; SVM, support vector machine.

research focus. Gadolinium-enhanced inner ear MRI remains the only method for confirming EH; however, its clinical utility is limited by invasiveness, procedural complexity, and substantial time requirements. AI has further enhanced the practicality of this imaging modality. As mentioned above, the deep learning model *INHEARIT* was developed to automatically quantify the EH ratio. It can reliably segment the vestibule and cochlea and calculate the proportion of hydrops volume relative to the total inner ear volume, demonstrating near-expert agreement with manual measurements.⁷ This tool may facilitate longitudinal follow-up by enabling quantitative monitoring of EH changes and treatment response. In addition, ML models based on pure-tone audiometry have been explored for automated MD diagnosis and EH prediction. One study showed that a Light Gradient Boosting Machine (LightGBM) model trained on multiple audiometric features achieved favorable performance in diagnosing MD, with an accuracy of 87% and a specificity of 90%, comparable to that of experienced otolaryngologists. The model also demonstrated potential for predicting EH subtypes, reaching an accuracy of 78%. Feature importance analysis indicated that the standard deviation of overall frequency thresholds, the mean low-frequency hearing level, and the audiogram peak contributed most to model performance. These findings suggest that pure-tone audiometry combined with ML may serve as an efficient and low-cost screening tool for MD.³²

5.3. AI applications in orthostatic intolerance

Orthostatic intolerance (OI) is a common cause of positional dizziness/vertigo in clinical practice. Its core feature is autonomic dysregulation, leading to symptoms such as dizziness, blurred vision or near-blackout, unsteadiness, and even transient loss of consciousness when changing from a supine to an upright posture. Gilmore et al.³³ applied ML-based classification to patients with OI to explore the feasibility of AI-assisted evaluation of posture-related dizziness/vertigo. The study analyzed dynamic heart rate and blood pressure data collected during head-up tilt testing in 186 participants, including individuals with postural orthostatic tachycardia syndrome, vasodepressor type, cardioinhibitory type, mixed type, and healthy controls. Using RF to extract cardiovascular response features at key time points, the model achieved

a classification accuracy exceeding 93%, effectively distinguishing different types of orthostatic regulatory abnormalities and identifying subtypes within mixed presentations. In addition, K-means clustering revealed distinct autonomic regulation patterns even among healthy participants. These findings suggest that AI can mine dynamic features related to vestibular and autonomic function from routine physiological signals, providing a novel technical pathway for intelligent subtyping and decision support in the diagnosis of orthostatic dizziness/vertigo.

5.4. AI applications in sudden sensorineural hearing loss with vertigo

ML models can assist in identifying high-risk patients with sudden sensorineural hearing loss (SSNHL) accompanied by vertigo and support expert decision-making, thereby improving prognostic assessment. Lin et al.³⁴ investigated the association between high-amplitude squared wavelet coherence derived from vHIT and hearing outcomes in patients with SSNHL accompanied by vertigo. By integrating wavelet coherence analysis with CNN- and SVM-based classification and prediction, the study found that high-coherence frequency in the posterior semicircular canal was an independent positive predictor of hearing recovery. Moreover, a hybrid CNN + SVM model using ResNet-50 for feature extraction achieved the highest accuracy in hearing outcome classification, with a mean accuracy of 91.6%. This approach outperformed conventional time-series analysis and CNN-only classification, providing a more stable and reliable tool for predicting hearing prognosis in clinical practice. To facilitate comparison of the applications, input data, performance, and limitations of different AI technologies in dizziness/vertigo disorders, the studies discussed above are summarized in Table 1.

6. Limitations and challenges

6.1. Data Quality and Standardization

Insufficient data quality and standardization are core factors limiting model performance and generalizability. Dizziness/vertigo disorders are characterized by complex etiologies, subjective symptom descriptions, and intermittent attacks. Across studies, substantial

Table 1
Summary of AI applications in dizziness/vertigo disorders.

Disorder / application scenario	AI model or current technical application	Key performance and applicable settings	Input data modalities	Major limitations
Stroke screening in AVS ¹⁴	LSTM model directly analyzing raw vHIT eye- and head-velocity time-series data	Stroke diagnostic accuracy 87.9%, sensitivity 87.7%, specificity 88.4%; applicable for central-cause screening in emergency AVS	Raw vHIT time-series data	Small sample size; needs integration with HINTS, imaging, and emergency workflows
Differentiation between central and peripheral vertigo ²⁰	The geometric deep learning model MultiGMC integrates clinical information	AUC 0.96, outperforming HINTS and ABCD2; maintained performance in atypical AVS without spontaneous nystagmus	Medical history, physical signs, oculomotor findings, and clinical variables	Mainly focused on stroke versus vestibular neuritis; performance in more complex real-world emergency populations remains uncertain
Differentiation between vestibular neuritis and posterior circulation stroke ²¹	Time-series ML algorithms based on vHIT data, including Rocket, Arsenal, catch22, and random interval classifiers	Rocket achieved 87.8% accuracy in an external validation cohort; applicable for rapid emergency differentiation	vHIT eye- and head-velocity time-series data	Requires standardized vHIT acquisition; cross-device generalizability remains unclear
Multiclass diagnosis of vestibular disorders ²²	CatBoost and other gradient-boosting models combining clinical features and expert-informed rules for BPPV, VM, MD, HOD, PPPD, and related disorders	Overall accuracy 88.4%; applicable for outpatient differential diagnosis of vertigo disorders	Medical history, symptoms, signs, and structured examination data	Depends on high-quality structured data; remains challenging in patients with comorbidities or multiple diagnoses
Multiclass diagnosis of vestibular disorders ²³	Machine-learning models based on multivariable analysis of a large vertigo patient database	Useful for classifying functional dizziness, MD, VM, and related vestibular disorders; applicable to outpatient decision support	Structured clinical database	Performance depends on database quality, diagnostic labeling, and representativeness of included patients
BPPV risk-factor identification and diagnostic prediction ²⁶	LASSO regression and random forest models incorporating inflammatory and metabolic indicators	Final multivariable model achieved an AUC of 0.927; useful for identifying potential risk factors and assisting BPPV prediction	Clinical variables, laboratory inflammatory and metabolic markers	Association-based findings; causal mechanisms remain uncertain; external validation needed
Gait stability assessment in BPPV ¹⁹	SVM model using gait parameters collected by wearable accelerometers to distinguish BPPV from healthy controls and predict DHI severity	AUC 0.78 for distinguishing BPPV from healthy controls; AUCs of 0.83, 0.85, and 0.96 for mild, moderate, and severe DHI groups	Wearable sensor data and gait parameters	Gait is influenced by age, comorbidities, and environment; cannot replace positional testing
Nystagmus detection in BPPV ²⁷	CNN- and LSTM-based models analyzing VNG data for eye movement and nystagmus detection	1D-CNN achieved accuracy 91.02% and F1-score 92.68%; applicable for automated BPPV nystagmus recognition	VNG eye movement data and pupil tracking	Sensitive to eye-movement recording quality; performance in complex or atypical BPPV requires improvement
Multimodal diagnosis of BPPV ²⁸	Autoencoder and cross-attention model integrating nystagmus video features with head-position information	Mean accuracy 81.7%; applicable for BPPV subtype assessment	Nystagmus videos and head-position data	Accuracy needs further improvement; limited training scale and insufficient external validation
Prediction of BPPV canal subtype ²⁹	Machine-learning model using DHI-related information to predict the affected semicircular canal subtype	May assist in preliminary subtype prediction; applicable as a supplementary screening tool	DHI questionnaire-derived clinical information	Predictive value is limited; cannot replace positional nystagmus testing
Intelligent repositioning for BPPV ³⁰	SRM-IV diagnostic and treatment system using AI to identify the affected side and canal and select the repositioning maneuver	Sensitivity 92.3% and specificity 89.7% in refractory BPPV; applicable for complex or refractory BPPV	Real-time nystagmus videos, head-position information, and chair-position data	Requires specialized equipment; cost, accessibility, and multicenter validation remain limited
Imaging-assisted diagnosis of MD ⁶	Radiomics-based multilayer perceptron model extracting texture features from routine T2-weighted inner ear MRI to distinguish MD from healthy controls	Accuracy 82%, sensitivity 83%, specificity 82% in an independent test set; applicable for noninvasive adjunctive diagnosis of MD	Inner ear MRI texture features	Limited sample size; mainly retrospective design; variability in MRI scanners and acquisition protocols may affect generalizability
Quantification of endolymphatic hydrops in MD ⁷	CNN-based automatic segmentation of the cochlea and vestibule to calculate the EH ratio	Excellent agreement with manual measurements by neurotologists, with an ICC of 0.971; applicable for objective EH quantification and follow-up monitoring	Gadolinium-enhanced inner ear MRI	Requires high-quality enhanced MRI; limited accessibility and procedural complexity; further external validation needed
Vestibular neuritis detection ⁹	SVM classification based on VNG-derived pupil tracking and nystagmus features	Accuracy up to 94%; applicable for automated recognition of vestibular neuritis	VNG videos, nystagmus frequency, fast- and slow-phase velocity	Small sample size; model performance depends on video quality and labeling consistency
Differentiation between MD and VM ¹⁵	AdaBoost and random forest models integrating medical history, nystagmus, audiological data, and vestibular function tests	Accuracy up to 97.8% when full examination data were available; accuracy > 92% using history-based information alone	Medical history, audiometry, nystagmus, and vestibular function tests	Mainly derived from specialized neurotology settings; performance in primary care or general outpatient clinics requires validation
Automated MD diagnosis and EH prediction ³¹	LightGBM model using pure-tone audiometric features for MD diagnosis and EH subtype prediction	Accuracy 87% and specificity 90% for MD diagnosis; accuracy 78% for EH subtype prediction	Pure-tone audiometry and audiogram-derived features	More suitable as a screening tool; cannot replace clinical criteria or imaging evidence

(continued on next page)

Table 1 (continued)

Disorder / application scenario	AI model or current technical application	Key performance and applicable settings	Input data modalities	Major limitations
Classification of orthostatic intolerance ³²	Random forest and K-means models analyzing dynamic heart rate and blood pressure changes during head-up tilt testing	Classification accuracy > 93%; applicable for intelligent subtyping of orthostatic regulatory abnormalities	Head-up tilt test, heart-rate and blood-pressure time-series data	Requires standardized tilt-test protocols; integration with vertigo-specific diagnostic pathways remains limited
Prognostic prediction in SSNHL with vertigo ³³	Wavelet coherence analysis combined with CNN and SVM to predict hearing recovery	CNN + SVM achieved mean accuracy 91.6%; applicable for hearing-prognosis assessment in SSNHL with vertigo	vHIT signals and wavelet coherence features	Disease-specific application; prospective validation is needed to confirm its impact on treatment decisions
Post-concussion vestibular subtyping ²⁴	Unsupervised clustering using a multilayer self-organizing map algorithm	Identified two clinically distinct vestibular profiles; useful for exploring latent disease subtypes	Balance and vestibular diagnostic data	Exploratory analysis; small cohort; requires validation in independent populations
Vestibular migraine phenotyping ²⁵	Clustering analysis to characterize phenotypic heterogeneity in VM	Identified five clinical subgroups; useful for exploring biological and clinical heterogeneity	Clinical features of VM, including vertigo duration, migraine-associated symptoms, and cochlear symptoms	Exploratory subtype discovery; clinical implications and treatment relevance require further validation

Table footnote: AI, artificial intelligence; ML, machine learning; DL, deep learning; MD, Ménière's disease; EH, endolymphatic hydrops; MRI, magnetic resonance imaging; CNN, convolutional neural network; VNG, videonystagmography; SVM, support vector machine; LSTM, long short-term memory network; vHIT, video head impulse test; AVS, acute vestibular syndrome; HINTS, head impulse, nystagmus, test of skew; ABCD2, age, blood pressure, clinical features, duration of symptoms, and diabetes; COP, center of pressure; BPPV, benign paroxysmal positional vertigo; DHI, Dizziness Handicap Inventory; VM, vestibular migraine; HOD, hemodynamic orthostatic dizziness; PPPD, persistent postural-perceptual dizziness; SSNHL, sudden sensorineural hearing loss.

differences exist in inclusion criteria, diagnostic standards, examination protocols, and outcome measures, resulting in marked data heterogeneity. Moreover, most existing studies are single-center, small-sample, retrospective analyses, and their patient populations may differ from real-world clinical settings. This may lead to model overfitting and limit generalizability across primary care, emergency departments, and diverse patient populations. In addition, the annotation of multimodal data, including nystagmus videos, vestibular function tests, imaging data, and clinical text, is highly dependent on expert experience. Inter-rater variability may further affect the stability of model training and the reliability of model outputs.

6.2. Insufficient Model Interpretability and Trustworthiness

Insufficient model interpretability remains an important barrier to the clinical implementation of AI-assisted diagnosis. Although DL models can automatically extract features from complex signals and images, their internal decision-making processes are often regarded as “black boxes,” making it difficult for clinicians to identify the features on which model predictions are based. When AI-generated outputs are inconsistent with clinicians' experience or conventional test results, clinicians may be reluctant to trust and adopt model recommendations in the absence of explainable evidence. For dizziness/vertigo, diagnosis requires dynamic reasoning that integrates medical history, physical signs, oculomotor findings, and ancillary tests. Therefore, providing only classification results or risk scores may be insufficient for clinical needs. Future models should not only provide diagnostic predictions but also offer traceable feature contributions, key supporting evidence, and interpretable diagnostic reasoning pathways.

6.3. Barriers to Clinical Translation

Evidence supporting clinical translation remains insufficient. Most current AI models are trained and internally validated using retrospective datasets. Although some studies have performed external validation, prospective cohort studies, real-world implementation studies, and randomized controlled trials remain scarce. Therefore, existing findings are still insufficient to demonstrate that AI systems can consistently improve diagnostic accuracy, shorten diagnostic pathways, or improve patient outcomes in clinical practice. In addition, clinical deployment of AI tools also faces challenges related to workflow integration. For example, nystagmus video acquisition, vHIT data processing, automated imaging segmentation, and multimodal information entry all need to be integrated with existing hospital information systems, examination devices, and clinicians' decision-making workflows. If AI systems increase additional operational burden, or if their outputs cannot be delivered to clinicians in a timely and intuitive manner, their practical value in clinical settings will be limited.

6.4. Ethical and Legal Challenges

The development of AI models relies on large amounts of patient data, including sensitive information such as medical history, imaging data, eye movement videos, physiological signals, and laboratory indicators. Therefore, strict principles of data de-identification, authorized use, and secure storage must be followed. Although cross-center data sharing can improve model generalizability, it also increases the complexity of privacy protection and data governance. In addition, when AI-assisted diagnostic errors lead to misdiagnosis or delayed treatment, it remains unclear whether responsibility should be borne by model developers, healthcare institutions, device manufacturers, or clinicians. Therefore, when promoting the application of AI in dizziness/vertigo care, it is necessary to establish data standards, model evaluation frameworks, clinical regulatory pathways, and accountability mechanisms in parallel, so as to ensure that AI can serve clinical practice in a safe, reliable, and sustainable manner.

7. Summary and future perspectives

AI has demonstrated broad potential in the diagnosis and management of dizziness/vertigo disorders. Across medical imaging, vestibular function testing, gait monitoring, and structured or unstructured clinical data, AI has shown preliminary value in feature extraction, pattern recognition, etiological classification, and treatment outcome prediction. However, most current applications remain at an early stage of “pattern recognition” and are still far from replicating the complete clinical reasoning process of experienced clinicians. As discussed above, existing studies remain constrained by limitations in data quality, model generalizability, interpretability, and insufficient evidence for clinical translation. Therefore, future research should move beyond the pursuit of model performance alone and gradually shift toward the development of interpretable, validated, and workflow-integrated intelligent decision-support systems.

First, multimodal large models may represent an important future direction for intelligent dizziness/vertigo care. The diagnosis of dizziness/vertigo inherently relies on the integration of multisource information, including medical history, vestibular function tests, audiological profiles, and imaging findings. Traditional AI models are usually trained for a single task or a single modality, making it difficult for them to fully emulate the integrative reasoning process of clinicians. In contrast, multimodal large models can perform joint reasoning across heterogeneous data types, enabling more comprehensive assessment in settings closer to real-world clinical practice. At the architectural level, such models may include vision–language models capable of simultaneously processing nystagmus videos and free-text medical records, as well as Transformer-based frameworks that integrate eye-movement time series, audiological features, and imaging biomarkers into a shared representation space. In principle, these models could not only perform disease classification but also generate structured diagnostic reports, highlight examination findings, and recommend targeted additional tests, thereby more closely approximating the comprehensive reasoning process of experienced specialists.

Second, federated learning provides a feasible strategy to address the conflict between multicenter data sharing and privacy protection. The generalizability of AI models for dizziness/vertigo depends on large-scale, multicenter, high-quality data. However, substantial differences exist across hospitals in data formats, examination devices, diagnostic criteria, and privacy governance, and direct pooling of raw data often faces ethical, legal, and technical barriers. Federated learning enables multiple centers to collaboratively train models without sharing raw patient data, exchanging only model parameters or gradient information. This approach can improve the scale and population representativeness of training data while preserving patient privacy. In the future, federated learning-based dizziness/vertigo databases may integrate data from different regions, hospital levels, and device sources, thereby facilitating model validation and optimization across emergency departments, outpatient clinics, primary care settings, and specialized centers.

Third, AI applications in dizziness/vertigo should further expand from diagnostic assistance to treatment decision support. Most current studies focus on disease classification, nystagmus recognition, stroke screening, or imaging segmentation. However, the real-world management of patients with dizziness/vertigo requires not only diagnostic clarification but also individualized treatment and follow-up strategies. In BPPV, AI may assist in identifying the affected semicircular canal and side based on nystagmus direction, duration, head-position information, and previous responses to repositioning maneuvers, and may recommend the most appropriate canalith repositioning procedure. For patients with recurrent or refractory BPPV, AI models may also predict the number of required repositioning sessions, recurrence risk, and follow-up frequency. In chronic or recurrent disorders such as MD and VM, AI may integrate hearing changes, attack frequency, triggers, treatment responses, and quality-of-life scales to assess disease activity,

predict recurrence risk, and provide recommendations for medication adjustment, lifestyle intervention, and rehabilitation training. For patients with acute vestibular syndrome, AI may not only assist in identifying central causes but also support triage, referral, and further testing decisions by integrating stroke risk, imaging findings, and emergency-care resources.

In addition, AI may facilitate the transition of dizziness/vertigo management from hospital-based episodic care to continuous, remote, and individualized care. With the development of smartphones, wearable devices, and related tools, vertigo episodes, postural and gait stability, and rehabilitation adherence can be continuously collected outside the hospital. AI can analyze these dynamic data to support early warning, treatment evaluation, and rehabilitation adjustment. In the future, human–AI collaboration should become the core application model for AI systems in dizziness/vertigo care, with clinicians retaining leadership in diagnostic and therapeutic decision-making while AI provides candidate diagnoses, risk alerts, test recommendations, and treatment references. Through the integration of multimodal large models, federated learning, interpretable modeling, and clinical decision-support systems, AI is expected to evolve from an “auxiliary recognition tool” into an “intelligent assistant” that supports the entire pathway of diagnosis, treatment, rehabilitation, and long-term management, ultimately contributing to a more efficient, safe, and precise healthcare system for dizziness/vertigo.

Consent for publication

All authors have read and approved the final version of the manuscript and consent to its submission for publication. We confirm that this work is original and has not been published elsewhere, nor is it currently under consideration for publication elsewhere.

Ethics statement

Ethics approval was not required by the ethics committee of the First Hospital of Hebei Medical University, as this review article does not contain any studies with human participants or animals performed by any of the authors.

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Data availability

This article is a review, and no experimental data are available.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-assisted technologies

During the research and preparation of this manuscript, the authors did not employ any generative AI or AI-assisted technologies for data analysis, interpretation, figure generation, or text composition. The authors are solely responsible for the integrity and originality of all content presented.

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